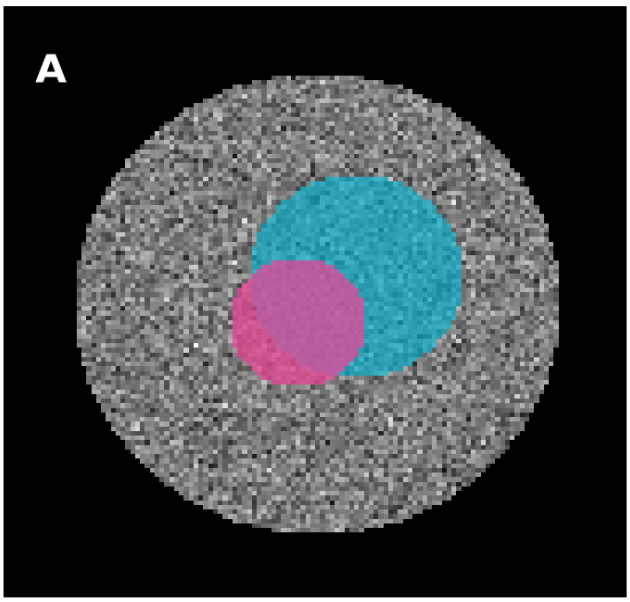
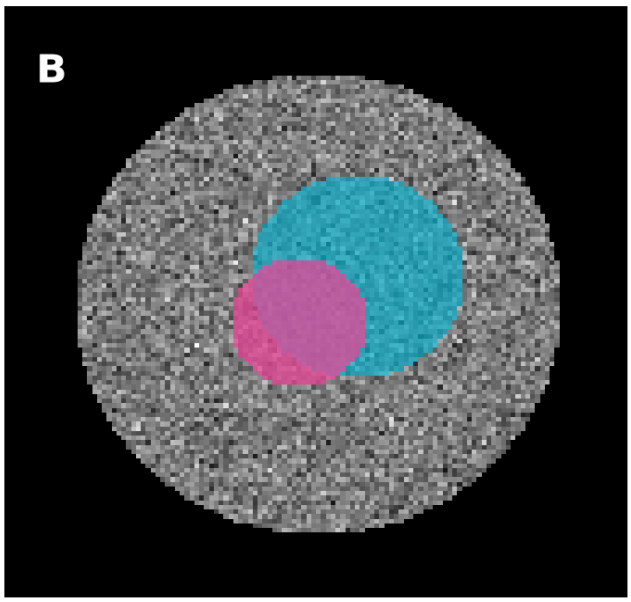
PhysioRANO NEURO-ONCOLOGY RADIOLOGY REPORT

Axial T1c (Post-Contrast)



Sagittal T2-FLAIR (Edema & Core)



CLINICAL HISTORY:

49-year-old Male presenting for post-radiotherapy Glioblastoma (GBM) evaluation at **20 days** post-treatment.
• Molecular Profile: IDH1/2 **negative** | MGMT Promoter **negative**.
• Current Status: Routine follow-up scan monitoring, mild headaches.

FINDINGS:

BRAIN PARENCHYMA & 3D TUMOR SUB-REGIONS:
Multi-modal 3D MRI quantitative volumetric analysis demonstrates:
• **Whole Tumor (WT) Volume: 24.50 cm³** (includes surrounding T2-FLAIR signal hyperintensity).
• **Tumor Core (TC) Volume: 12.20 cm³**.
• **Contrast-Enhancing Tumor (ET) Volume: 18.54 cm³**.

BIOPHYSICAL TUMOR GROWTH TRAJECTORY:
Biophysical neural network modeling evaluates the spatial invasion and proliferation dynamics:
• **Tissue Diffusion Speed:** Low spatial diffusion coefficient (**1.0260 mm²/day**).
• **Cell Proliferation Rate:** Modest cell proliferation rate (**0.7234 day■¹**).
• **Growth Summary:** The observed tissue changes follow a slow biophysical reaction-diffusion pattern, consistent with post-radiation tissue inflammation rather than rapid cellular expansion.

MULTIMODAL RECURRENCE & PSEUDOPROGRESSION ANALYSIS:
Integrated machine learning analysis evaluating clinical biomarkers, timeline post-treatment, and biophysical growth dynamics yields:
• **Probability of Pseudoprogression (Treatment Effect): 30.9%**
• Probability of True Tumor Progression: 58.2%
• Probability of Stable Disease: 10.9%
• **Key Contributing Factors:** IDH mutation status, favorable MGMT methylation, and timing within 12 weeks post-radiotherapy strongly favor treatment-induced tissue reaction.

RANO 2.0 CLINICAL RESPONSE EVALUATION:
Comparison with baseline MRI scans shows an enhancing volume shift from **12.00 cm³** to **18.54 cm³** (a **+52.7%** volumetric change).
Under standard RANO 2.0 guidelines within the first 12 weeks post-radiotherapy, an enhancing expansion supported by high treatment-effect probability is classified as **Progressive Disease (PD)**.

3D CONDITIONAL GAN (cGAN) LONGITUDINAL FORECAST (+90 DAYS):
Generative biophysical 3D synthesis conditioned on spatial diffusion and proliferation predicts:
• **Projected Whole Tumor Volume: 480.28 cm³** (+133.3% trajectory shift).
• **Projected Enhancing Tumor Volume: 263.72 cm³** (+133.3% trajectory shift).
• **Clinical Trajectory Interpretation:** Projected morphological progression aligns with slow inflammatory stabilization under current adjuvant regimen.

IMPRESSION:

The enhancing volume shift (+52.7%) observed at 20 days post-radiotherapy in an IDH-mutated, MGMT-methylated glioblastoma patient is clinically consistent with **TRUE PROGRESSION (Radiation Necrosis / Treatment Effect)** rather than true disease recurrence.

CLINICAL RECOMMENDATION: Significant volumetric expansion (>= 25%) or new lesion indicative of true tumor progression.